

tests/stress/ proxy_forward_stress.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Authored + parse-clean. Runs live against the running Squid
HTTP forward proxy (localhost:53128); honest SKIP (§11.4.3) when
the proxy or the target endpoint is unreachable.

Companion (§11.4.18) to the in-source documentation block
at the top of the script.

Overview

§11.4.85 / §11.4.169 **stress** suite for the LIVE HTTP forward proxy
(Squid, localhost:53128). It drives **N ≥ 100 sequential** HTTPS-
CONNECT requests through the proxy **plus a concurrent burst**
of ≥ 10 parallel requests, captures each request's `{http_code}`
and `{time_total}` to its **own file** (so a verdict never rests on a
background job's exit status — the B3 bluff fix), and records a p50/
p95/p99 latency distribution to a captured `latency.txt`. It **PASSes**
only when **every** proxied request succeeded (204/200) with no
deadlock (every request returned within `--max-time`) and no crash.
A representative plain-HTTP probe captures the Squid `Via: header`
— hard client-side proof the bytes transited `proxy-squid`. Every
PASS cites captured evidence (§11.4.69); a metadata-only PASS is
refused.

Prerequisites

- Committed library `tests/lib/evidence.sh` (sourced — provides
`_code_in`, `port_is_listening`, `proxy_conn_verdict`,
`ab_pass_with_evidence`, `ab_skip_with_reason`).
- `curl`, `awk`, `sort`, `grep`, POSIX `sh`, `date`.
- A running Squid forward proxy on `HTTP_PROXY_PORT` for a non-
SKIP run.
- Write access to `qa-results/` (gitignored).

Usage examples

- Default run against localhost:53128: `bash tests/stress/proxy_forward_stress.sh`
- Conductor invocation under host-safety caps (§12.6):
`GOMAXPROCS=2 nice -n 19 ionice -c 3 bash tests/stress/proxy_forward_stress.sh`
- Custom load / target: `STRESS_SEQ=100 STRESS_CONC=10 STRESS_TARGET=https://www.gstatic.com/generate_204 bash tests/stress/proxy_forward_stress.sh`

Env knobs: `HTTP_PROXY_URL` (default `http://localhost:53128`), `HTTP_PROXY_PORT` (default 53128), `STRESS_TARGET` (default `https://www.gstatic.com/generate_204`), `STRESS_EXPECT` (default 204 200), `STRESS_VIA_TARGET` (default `http://www.gstatic.com/generate_204`), `STRESS_SEQ` (default 100), `STRESS_CONC` (default 10), `CURL_MAX_TIME` (default 20), `STRESS_EVIDENCE_DIR` (default `qa-results/stress/proxy_forward_<ts>`).

Edge cases

- **Every proxied request succeeds** → PASS, citing `latency.txt`. Exit 0.
- **Some requests dropped BUT the target is reachable directly** → FAIL, a real proxy defect / possible deadlock (§11.4.68 — a listening-but-dropping proxy can never fail-open to SKIP). Exit 1.
- **Requests dropped AND the target is unreachable directly too** → SKIP (`network_unreachable_external`) — a third-party outage is not a proxy defect (§11.4.1). Exit 3.
- **Proxy port not listening + no direct signal** → SKIP (`topology_unsupported`). Exit 3.
- **Deadlock guard**: no request can exceed `CURL_MAX_TIME`; a hung request reports 000 and counts as a failure — never hangs the suite.

Internal behaviour

- `#!/usr/bin/env bash`, POSIX-clean body (`sh -n + bash -n`, §11.4.67).
- Step 0 Via corroboration probe (plain HTTP) → `via_probe_headers.txt`.
- Sequential loop → `seq.N.code`; concurrent burst → `conc.N.code / conc.N.time` (bounded by `STRESS_CONC`); latency samples sorted numerically; nearest-rank p50/p95/p99 + min/max/mean computed in awk → `latency.txt`. Summary → `stress.evidence`.
- `trap ... EXIT INT TERM` reaps stray workers and removes the scratch dir (§11.4.14).

- Resources: shell + curl only, capped concurrency, well under the §12.6 60% host-memory ceiling.

Related

- tests/lib/evidence.sh — sourced anti-bluff evidence library.
- challenges/scripts/proxy_forward_http_challenge.sh — the functional forward Challenge this stress suite complements.
- tests/chaos/proxy_restart_recovery.sh, tests/security/proxy_acl_security.sh — sibling chaos + security suites.
- Constitution §11.4.85 / §11.4.169 / §11.4.69 / §11.4.1 / §11.4.68 / §11.4.50 / §11.4.107 / §12.6.

Last verified

2026-07-01 — authored; sh -n + bash -n parse-clean. Live sustained / concurrent load is exercised by the conductor against the running proxy.